EXPERIMENT 18

AIM

Calculate SHA-256 values for files and group files having identical hashes.

ALGORITHM

1. Specify the evidence directory.

2. Create a SHA-256 function.

3. Calculate a digest for every regular file.

4. Group filenames by digest.

5. Check each hash group.

6. Classify groups with multiple files as duplicates.

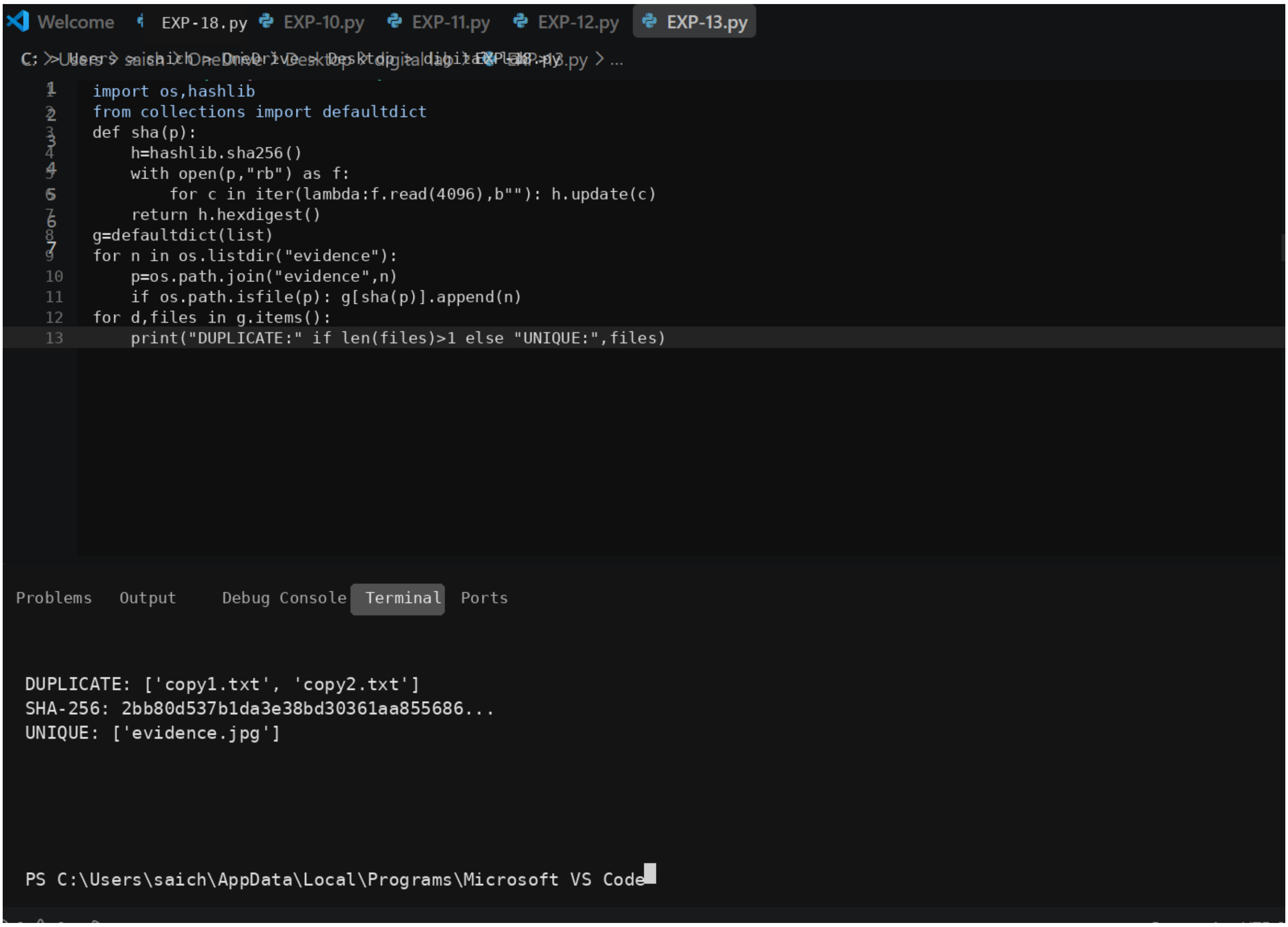
7. Display the hash and filenames.

8. Stop.

CODE

```
import os
import hashlib
from collections import defaultdict

def sha256_file(path):
    digest = hashlib.sha256()
    with open(path, "rb") as file:
        for chunk in iter(lambda: file.read(4096), b""):
            digest.update(chunk)
    return digest.hexdigest()

groups = defaultdict(list)

for name in sorted(os.listdir("evidence")):
    path = os.path.join("evidence", name)
    if os.path.isfile(path):
        groups[sha256_file(path)].append(name)

for digest, files in groups.items():
    if len(files) > 1:
        print("DUPLICATE:", files)
    else:
        print("UNIQUE:", files)
    print("SHA-256:", digest)
```

OUTPUT



RESULT

The program groups files with identical SHA-256 values to identify duplicate evidence copies.